ArthaSync

Unified Commerce Intelligence

Retail — Unified Commerce | Industry Theme: Retail

"A king should ensure that all trade routes report accurate prices and stocks — so no merchant profits from false information while another suffers from ignorance."

— Chanakya, Arthashastra, Book II | 321 BC

Business Plan (1/2)

WHY

Explain the Problem

Problem Description & Business Scenario	Indian retail is facing a fragmentation crisis. With over 63 million SMEs operating across physical stores, websites, and mobile apps simultaneously, the lack of unified inventory and pricing synchronisation costs each retail SME an estimated **Rs.2.4 lakh annually** (Deloitte India, 2024). When a customer places an order online and the store staff discover the item is out of stock after order confirmation — because POS, web, and app each maintain separate stock counters — the outcome is a cancelled order, a 1-star review, and a permanently lost customer. The Indian e-commerce market is projected to reach **Rs.10.6 lakh crore by 2026** (NASSCOM). Yet fewer than 5% of retail SMEs have any form of unified commerce infrastructure. Manual reconciliation consumes 6 hours of staff time per store per week. Overselling and price inconsistencies across channels account for 12–15% of revenue leakage. This is not a technology problem that requires expensive enterprise ERP systems — it is a data synchronisation and intelligence problem that can be solved with modern AI agents.
Problem Scope	Scope is defined around Indian omnichannel retail SMEs operating 2–5 sales channels concurrently (e-commerce website, mobile app, and physical POS store). The solution addresses: (1) real-time inventory synchronisation across channels, (2) automated price conflict detection, (3) oversell risk identification, and (4) AI-generated actionable manager briefings. **Out of scope:** upstream supply chain management, ERP integration, international markets, and warehouse management systems.
Target Users / Stakeholders	**Primary — Retail Store Owners / Business Owners:** Need a single dashboard that shows the real health of inventory across all channels without hiring a data analyst. **Secondary — Store Managers:** Need plain-language daily briefings with specific action items (e.g. "Fix Running Shoes price on app — Rs.150 lower than web") without interpreting raw data. **Tertiary — E-Commerce Channel Managers:** Need conflict alerts in real time to prevent overselling events before they trigger customer complaints. **Cognizant Enterprise Clients:** Large omnichannel retailers (Big Bazaar, Reliance Retail, D-Mart franchisees) who need this at scale — a direct commercial opportunity for Cognizant.

HOW

Explain the Solve

Solution Overview	**ArthaSync** is an AI-powered Unified Commerce Intelligence platform that synchronises inventory, pricing, and promotional data across all retail channels in real time — and uses a 3-agent LangGraph AI pipeline to automatically detect conflicts and generate plain-English actionable briefs for store managers. The platform has three core modules: **1. Sync Agent:** Polls web store API, mobile app API, and POS webhook every 60 seconds. Normalises all three data formats into a single unified PostgreSQL snapshot. **2. Conflict Agent:** Scans the snapshot and flags every price mismatch where delta > Rs.20 across channels — a configurable threshold set at approximately 1.3% of the average SKU price (Rs.1,499), below the threshold at which customers typically comparison-shop across channels (McKinsey Consumer Behaviour Report, 2023). Also flags oversell risk when POS stock = 0 while online channels show stock > 10 units. **3. Insight Agent:** An LLM (AWS Bedrock Claude Haiku) converts the conflict data into a 3–5 bullet daily manager brief in plain English with specific numbers and clear actions. A React dashboard provides real-time channel overview cards, colour-coded conflict alerts, and the AI insight feed — accessible from any device.
Technical Details	**Programming Languages:** Python 3.11, JavaScript (TypeScript-ready) **Frontend:** React 18 with Axios polling (5-second refresh), CSS Modules **Backend / API:** FastAPI (Python) with SQLAlchemy ORM, CORS middleware, and background task processing **AI Agent Framework:** LangGraph (StateGraph) — explicitly listed in Cognizant guidelines **LLM Provider:** AWS Bedrock (Claude Haiku model) — runs entirely within AWS, zero external dependencies **Database:** PostgreSQL (AWS RDS db.t3.micro) with JSON columns for conflict metadata **Cloud Infrastructure:** AWS EC2 t2.micro (FastAPI server), AWS Lambda + EventBridge (60-second pipeline scheduler), AWS S3 + CloudFront (React hosting), AWS API Gateway (routing) **Agent Orchestration:** LangChain (model interface layer) + LangGraph (StateGraph pipeline) with typed state (CommerceState TypedDict) **Resilience & Failure Handling:** The pipeline implements exponential backoff on channel API failures, AWS SQS dead-letter queues on Lambda for failed sync events, and a cached last-known-good state pattern to prevent dashboard blackouts during channel outages. Duplicate POS webhook events are deduplicated via idempotency keys on the FastAPI endpoint. **Prior Platform Credibility:** Team lead Hussein Petladwala has built and shipped DigiVyapar — a *live, production AI automation platform* for Indian SMEs, integrating multi-step LLM workflow agents, invoice OCR processing, and real-time marketing content generation. ArthaSync is a direct architectural evolution of this platform applied to the retail commerce synchronisation domain. The team brings end-to-end production experience from LLM API integration to user-facing product delivery.
Innovation	**1. Multi-Agent LangGraph Orchestration:** Rather than a single monolithic model, ArthaSync uses three specialised agents in a typed state graph — each with a single responsibility. This means the conflict logic (pure Python rules) and the language generation (LLM) are independently updatable, testable, and explainable — a significant architectural advantage over single-prompt or chain-based approaches. **2. LLM-Generated Natural Language Intelligence:** Existing retail dashboards show numbers. ArthaSync tells managers what to do. The Insight Agent converts raw conflict data into actionable plain-English briefs — eliminating the need for data literacy among store staff. **3. Historical Conceptual Framework (Arthashastra, 321 BC):** The platform is named after Chanakya's Arthashastra — the world's first unified commerce management system, which mandated cross-market price reporting to prevent information asymmetry. ArthaSync is the digital implementation of this 2300-year-old principle, contextualised for modern Indian retail. **4. Zero External Dependencies on AWS:** By using AWS Bedrock for LLM inference, the entire solution — data, agents, LLM, frontend — runs inside a single AWS environment with no external API calls, ensuring data privacy and compliance for retail client data.

Market Potential	**TAM — Total Addressable Market:** $4.8B globally for omnichannel retail software by 2028 (Gartner); Rs.10.6 lakh crore Indian e-commerce market in 2026 (NASSCOM) **SAM — Serviceable Addressable Market:** 3.15M Indian SMEs operating 2+ digital channels (5% of 63M total SMEs) × Rs.2.4L average annual sync loss = **Rs.75,600 crore addressable annual pain** (Deloitte India, 2024) **SOM — Serviceable Obtainable Market (Year 1):** 1,000 pilot stores across Pune, Mumbai, Bengaluru at Rs.42,000/year per store = **Rs.4.2 crore ARR in Year 1**. Scaling to 10,000 stores in Year 2 = Rs.42 crore ARR. 0.1% SAM capture in Year 3 = Rs.151 crore ARR. **Competitive Landscape:** No existing solution targets Indian SME omnichannel sync at this price point. Enterprise platforms (Salesforce Commerce Cloud, SAP Hybris) start at Rs.50L/year — 10× above SME budget. ArthaSync at Rs.42K/year is a **blue ocean segment**. **Growth Driver:** India digital retail penetration at 38% CAGR (NASSCOM 2024); omnichannel adoption accelerating in Tier 2 and Tier 3 cities post-pandemic.
Why These Technologies Are Appealing	**LangGraph:** Typed state graph keeps Sync, Conflict, and Insight agents decoupled — more explainable than LangChain single-chain. **FastAPI:** Async-native; handles concurrent multi-channel webhooks without blocking. **AWS Bedrock:** Enables LLM inference entirely within the Cognizant-provided AWS environment. No OpenAI or Anthropic API keys required — data never leaves AWS, ensuring client data privacy. **PostgreSQL:** ACID-compliant relational database with JSON support — provides both structured inventory tables and flexible conflict metadata storage. Scales to 10M+ SKUs on RDS. **React:** Polling (5s) gives real-time feel; enables parallel team development; upgradeable to WebSocket streaming.

WHAT *Value Proposition*	
Primary Benefits	**Business Value Delivered:** ArthaSync eliminates the primary causes of revenue leakage in omnichannel retail — price inconsistency and inventory inaccuracy across channels — through automated AI intelligence rather than manual processes. The result is a measurable, quantifiable improvement in both operational efficiency and customer experience.
Efficiency & Flexibility	• **60% reduction in conflict resolution time** — automated detection in under 60 seconds vs. avg 4–6 hour manual discovery cycle (projected estimate based on Gartner 2024 retail operations benchmark: manual cross-channel reconciliation averages 4.2 hours per incident; ArthaSync targets sub-60-second detection) • **6 hours/week saved** per store in manual reconciliation staff time • **Modular agent architecture:** each of the 3 agents (Sync, Conflict, Insight) is independently deployable and upgradeable without affecting others • Scales from a single-store retailer to a 500-store chain — application code unchanged; only infrastructure provisioning scales
Time & Cost Saving	• **ROI-positive within 6 weeks** for a 10-outlet retailer — derivation: Rs.2.4L annual loss eliminated ÷ Rs.42,000 annual cloud cost (Rs.3,500/month × 12) = **470% annual ROI**. Cloud cost breakdown: EC2 t2.micro Rs.1,200/mo + RDS db.t3.micro Rs.1,500/mo + Lambda + S3 + Bedrock inference Rs.800/mo = Rs.3,500/month total. • **15% revenue recovery** from elimination of overselling and price arbitrage losses • **Zero additional headcount** required — the Insight Agent replaces 2 hours of daily manual reporting per store manager • Deployed on AWS free tier during the competition phase; production cost estimated at Rs.3,500/month for a 10-store retailer on RDS + EC2 + Lambda

Scalability	• **Data layer:** PostgreSQL on AWS RDS scales horizontally from 10 to 10 million SKUs with no application code changes • **Agent layer:** LangGraph agents are stateless and can be deployed as parallel Lambda functions — 1 to 1,000 stores adds only infrastructure, not complexity • **Channel layer:** New channels (WhatsApp Commerce, Meesho, Amazon) are added as new sync_node connectors — existing conflict and insight logic remains unchanged • **LLM layer:** AWS Bedrock auto-scales inference; upgrading between available Bedrock model tiers requires only a single parameter change
Social Impact	• **SME empowerment:** Gives a small Pune retailer the same inventory intelligence capability that Reliance Retail has with a 200-person tech team — at Rs.3,500/month • **Waste reduction:** Eliminates overselling-driven logistics waste (return shipping, repackaging) contributing to lower carbon footprint per order • **Employment quality:** Frees store staff from 6 hours/week of tedious reconciliation, redirecting time to customer service and sales • **Digital inclusion:** Platform works in Hindi and regional languages via Bedrock multilingual LLM — accessible to non-English-speaking store owners across Tier 2/3 cities

Financials & Timelines | Business Plan (2/2)

INVESTMENTS

What does it take & how much does it cost to solve?

Cloud + AI (AWS)	EC2 t2.micro + RDS PostgreSQL + S3 + Lambda + CloudFront + API Gateway + Bedrock (Claude Haiku inference). AI cost: ~$0.001/pipeline run; 1,440 syncs/day = ~$1.44/day.	~Rs.3,500/month at production scale. Free tier during hackathon.
Tools & Frameworks	All open-source: LangGraph, LangChain, FastAPI, React, PostgreSQL. Process: replaces 6 hrs/week manual reconciliation per store.	Zero licensing cost. One-time dev effort; no recurring tool cost.
Team	Hackathon: 4 engineers x 24 hours. Production: estimated 3-week sprint for enterprise onboarding.	Production: 1 backend + 1 frontend engineer for ongoing maintenance.

RETURNS

Quantify the benefits & what if I don't solve?

ROI	Rs.2.4L loss eliminated vs Rs.42K/yr cloud cost. (240,000-42,000) / 42,000 = **470% annual ROI**. Payback: 7 weeks.	**Without ArthaSync:** Continue losing Rs.2.4L/yr per SME. Competitor with unified system wins customer loyalty permanently.
Revenue Growth & Customer	**15% revenue recovery** from eliminated overselling. For a Rs.50L/yr retailer = Rs.7.5L additional revenue. Zero "item unavailable" incidents; NPS improvement +18-25 pts.	**Without ArthaSync:** Price gaps exploited by competitors. Each oversell = 1-star review + permanent churn. Indians share bad experiences 3x more than good.
Margin, Staff & Operations	Rs.180/refund x 40 events/mo (KPMG India 2023) = **Rs.86,400/yr saved**. Staff time freed: 6 hrs/week. Manager brief replaces 2 hrs/day reconciliation.	**Without ArthaSync:** Refund costs compound at scale. Staff demotivation from manual reconciliation. Industry attrition already 35% annually.

TIMELINES

Time to realise benefits — from hackathon to production

Phase 1 — Agent & Backend Apr 12–20	**Build**	LangGraph 3-agent pipeline end-to-end. FastAPI endpoints live. PostgreSQL schema. Mock conflict data.	**8 days**
Phase 2 — Frontend & AWS Apr 21–30	**Integrate**	React dashboard live. AWS EC2 + RDS deployed. Lambda + EventBridge scheduler. CloudFront URL.	**10 days**
Phase 3 — Demo Hardening May 1–5	**Polish**	Full demo rehearsed. Backup video recorded. Conflict scenario scripted. Deck finalised.	**5 days**
Phase 4 — MVP Deployment May 6–7, 2026	**Deploy**	In-person hackathon at Cognizant Pune. Full MVP on Cognizant AWS. Live demo May 7.	**24 hrs**
Post-Hackathon May–Jun 2026	**Scale**	Onboard 3 pilot Pune SMEs. Integrate Petpooja/Ginesys POS APIs. Add WhatsApp Commerce. Target: 10 stores.	**30 days**